Professor-in-a-Box

A Personalized AI Tutor Powered by QLoRA and a Cloud Data Pipeline

Final ISA Presentation · DATA 298A

Fall 2025 – Final ISA Demo

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End-to-End Cloud Pipeline · QLoRA Fine-Tuning · Real-Time Tutor

Our goal was to build an AI tutor that understands student reasoning and explains concepts the way a human instructor does.

Project Vision: Why We Built This

Computer Science education often lacks personalized guidance for deep conceptual understanding. Complex topics like recursion and debugging require tailored explanations that static platforms cannot offer. We envisioned an AI tutor, 'Professor-in-a-Box', to provide real-time, personalized instruction, clarify concepts, and guide students through debugging with human-like reasoning.

Challenges in CS Learning

- Over-emphasis on problem-solving over conceptual depth
- Difficulty in getting personalized explanations
- Lack of support for guided debugging

AI Tutor Capabilities

- Real-time conceptual clarification
- Guided debugging pathways
- Reasoning-first explanations
- Scalable personalized tutoring

This semester, we completed the full ISA pipeline lifecycle: data ingestion → transformation → warehouse → monitoring → model evaluation → demo.

Semester Accomplishments (What We Did End-to-End)

01

Automated Cloud Pipeline (Airflow DAGs)

Fully automated Airflow pipeline for data ingestion and transformation.

02

Versioned Storage Layer (S3 Warehouse)

Designed a versioned S3 data warehouse with industry-standard practices.

03

Curated Instruction Dataset

Curated a 5,000-sample CS instruction dataset.

04

QLoRA Fine-Tuning (Qwen2.5-Coder-7B)

Fine-tuned Qwen2.5-Coder-7B with QLoRA on consumer hardware.

05

Real-Time Inference (<200ms)

Achieved real-time inference (<200ms) for interactive tutoring.

06

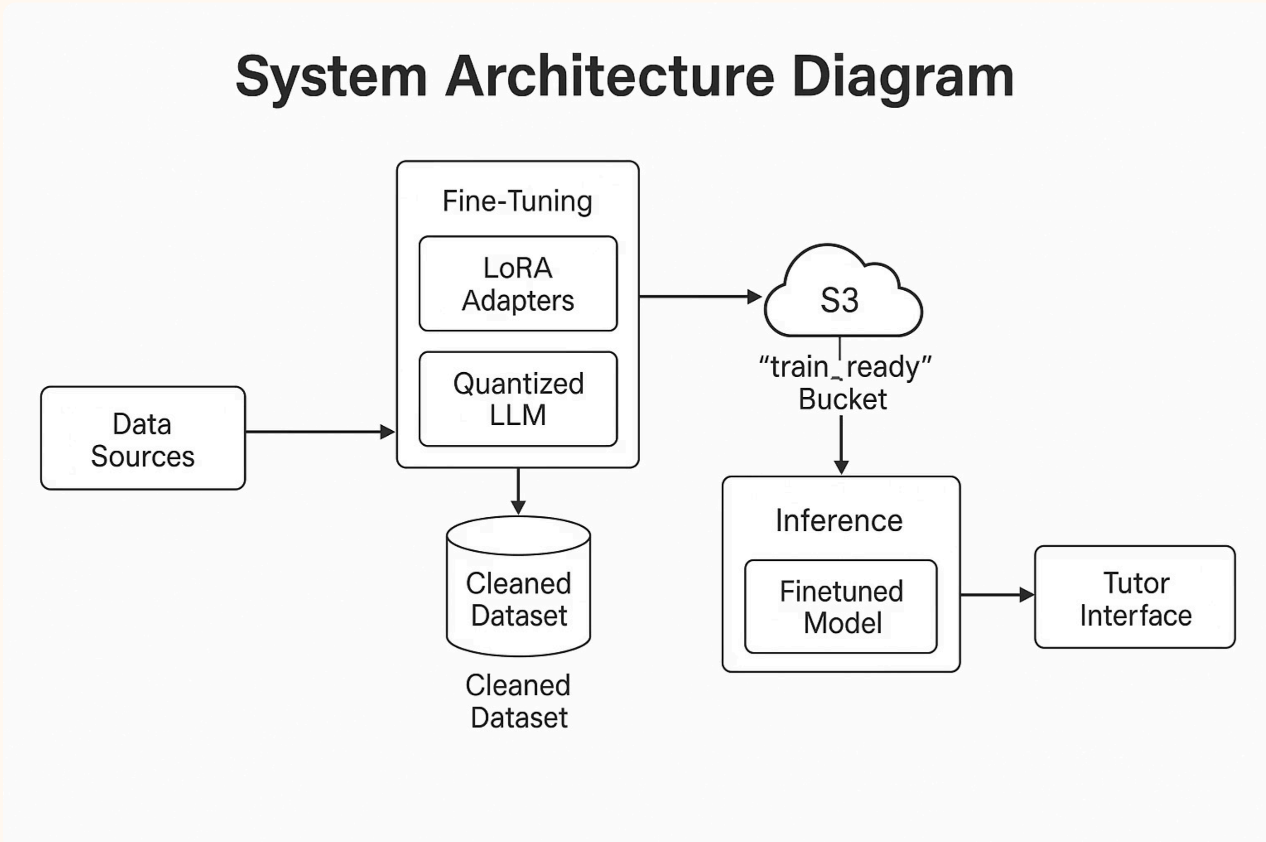
Complete ISA Demo (End-to-End)

Delivered a full ISA demo, covering the entire pipeline.

This architecture forms the foundation of our end-to-end ISA pipeline, connecting data ingestion through model serving.

End-to-End System Architecture

Our architecture integrates:



This design allows the system to be modular, scalable, and reproducible across training cycles.

Ingestion is fully automated, recoverable, and validated at every step.

Data Ingestion Pipeline (Airflow)

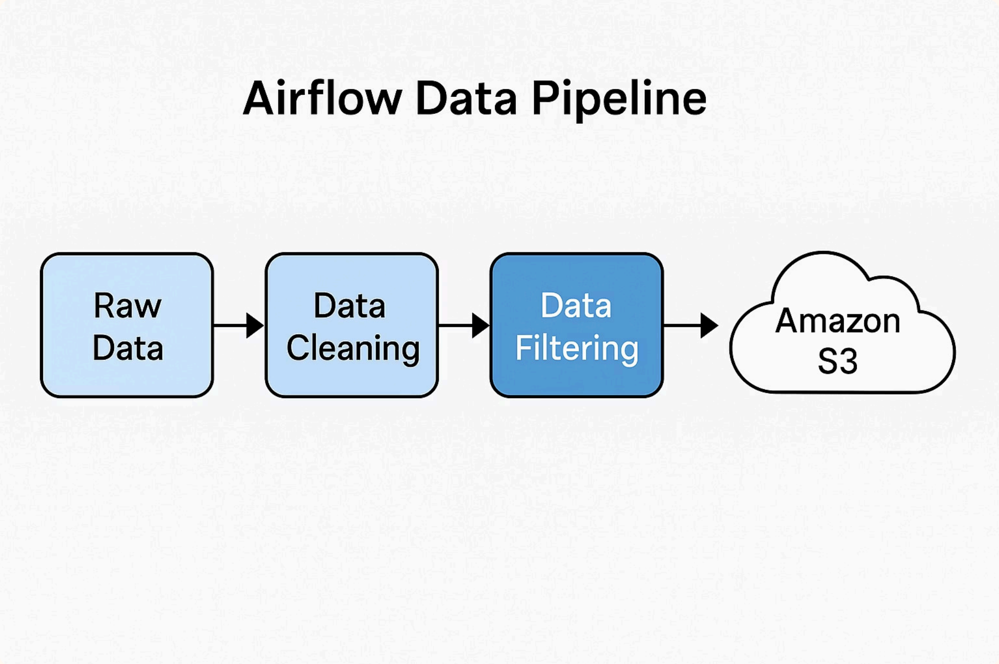
The ingestion pipeline is responsible for reliably importing large-scale dataset chunks from the Cosmopedia corpus.

Data is sourced from the Cosmopedia dataset, a large-scale corpus of instructional text.

Key Behaviors:

- Automated DAG triggers for raw dataset extraction
- Row-count validation per chunk
- Row-count validation prevents corrupted, partial, or incomplete ingestion files.
- Logging of ingestion events for traceability
- Upload of raw data to S3 (/raw/ folder)
- Support for retries and failure recovery

The Airflow DAG ensures that if any step fails, the pipeline can safely retry without data loss.



This ensures the pipeline remains robust, fault-tolerant, and fully reproducible.

Cleaning & Domain Filtering

Cosmopedia includes multi-domain content; filtering ensures the LLM learns only CS pedagogy.

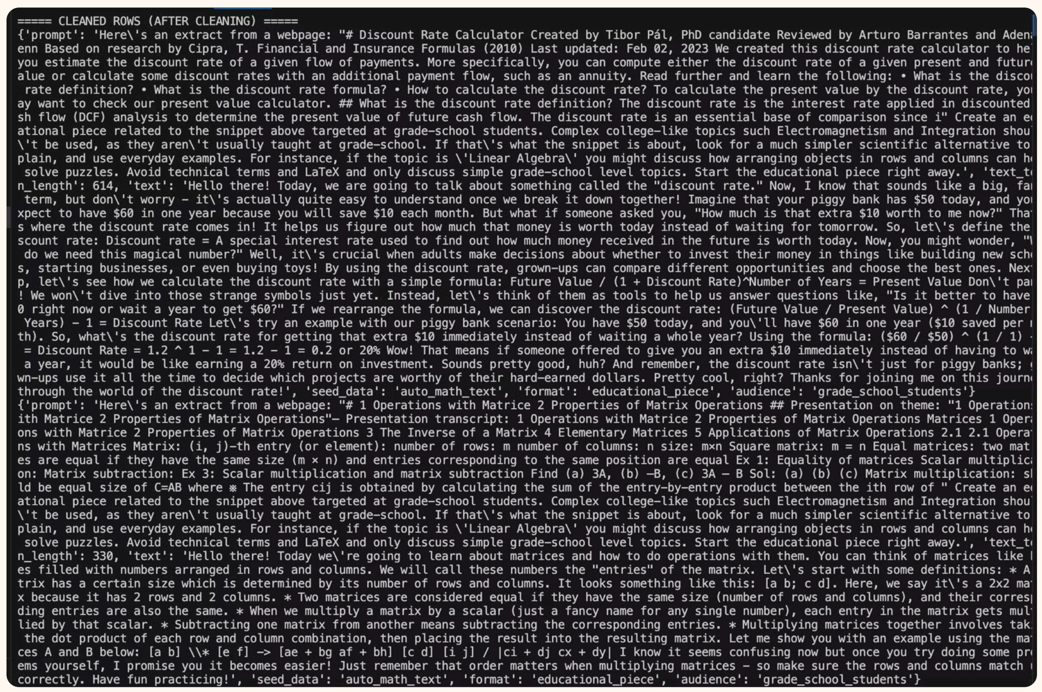
Raw data contained multiple issues: HTML noise, escaped characters, excessive formatting, and irrelevant non-CS content. Our cleaning pipeline performs:

- Text normalization & structural cleanup
- Removal of malformed/incomplete samples
- Domain filtering to isolate Computer Science content
- Semantic deduplication for dataset variety
- Quality enforcement ensures explanations remain pedagogical and suitable for educational use.

This stage transforms heterogeneous raw text into a high-quality tutoring dataset.

Before Cleaning

After Cleaning



After cleaning, we transform the text into a format suitable for LLM training.

Transformation to Model-Ready Format

After filtering, the cleaned dataset is transformed to prepare it for LLM training:

Transformation Pipeline:

- Parse cleaned text into instruction → response pairs
- Tokenize using Qwen2.5 tokenizer
- Generate token IDs and attention masks
- Serialize to JSONL (for training) and Parquet (for efficient streaming)
- Validate sample lengths and distributions

This structured format ensures consistent, well-formed inputs for QLoRA fine-tuning.

Model-Ready Data Structure:

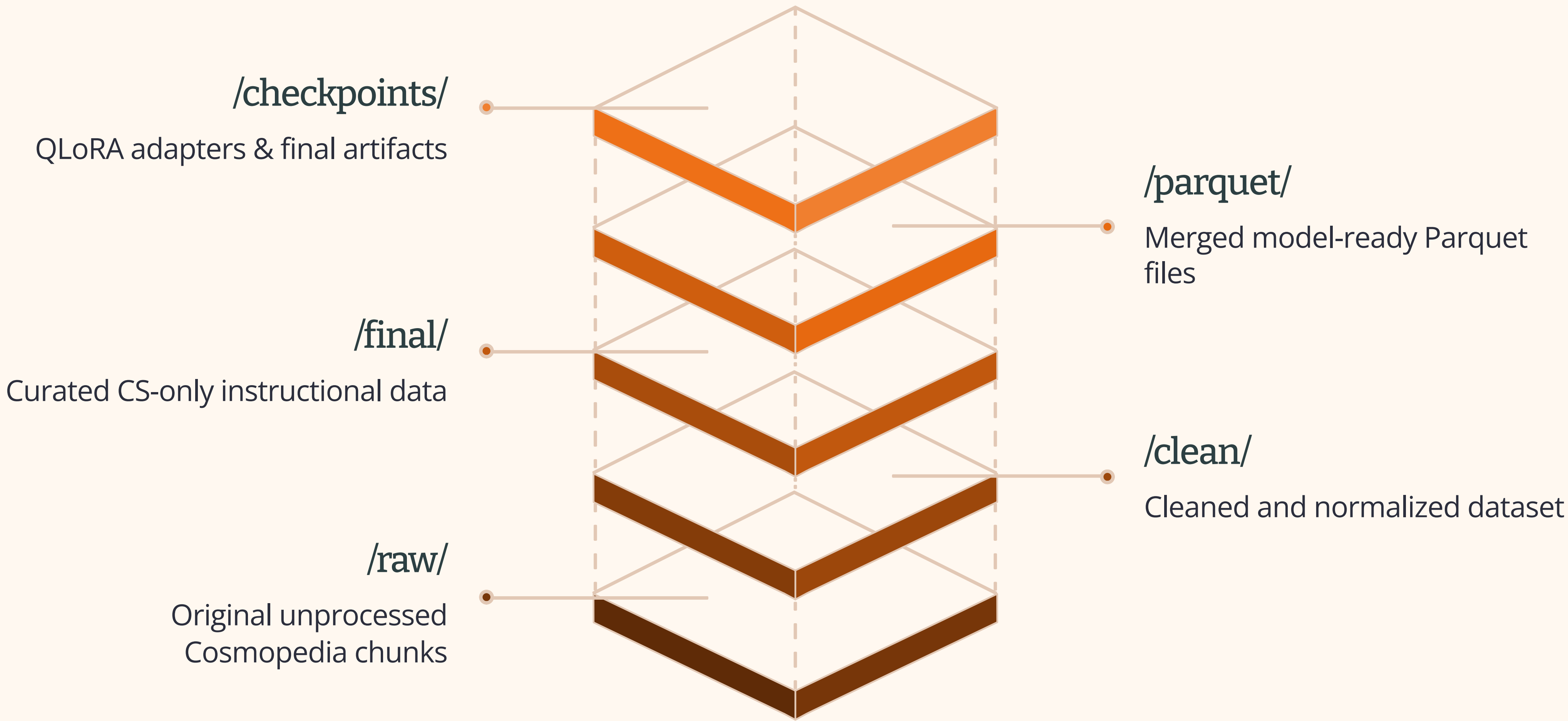
Field	Description	Example
instruction	Student-facing prompt	"What is recursion?"
response	Curated CS explanation	"Recursion is a technique where a function calls itself..."
token_count	Number of tokens	256
domain_tag	CS topic category	"algorithms"

The S3 layout ensures reproducibility, versioning, and simplifies training restarts.

Amazon S3 acts as our centralized storage system:

Each directory maintains version history, allowing rollback to previous data states if needed.

Bucket Layout:



Parquet format enables high-performance streaming during QLoRA training, reducing memory overhead.

This modular structure supports scalability, versioning, reproducibility, seamless retraining, and dataset expansion.

Monitoring & Pipeline Management

Robust monitoring and automated recovery are critical for production-scale AI pipelines.

To ensure reliability and auditability, our pipeline includes:

Airflow Monitoring: Visualizes task status, DAG health, execution history, and provides alerting

Automatic Retries: Handles S3 timeouts, transient failures, and network issues with exponential backoff

Log Inspection: Enables debugging of individual cleaning, transformation, and ingestion steps

Data Validation: Row-count checks at every stage detect anomalies and data quality issues early

Automated Row Count Comparison: Before/after each stage validates data integrity.

S3 Version Checks: Ensures safe, consistent storage and enables rollback if needed

This strengthens the pipeline's operational reliability, essential for real educational-scale AI workflows.

Airflow also provides alerting, retry logic, and comprehensive logs for debugging.

QLoRA Model Training

Qwen2.5-Coder-7B is optimized for code understanding and reasoning, making it ideal for CS education.

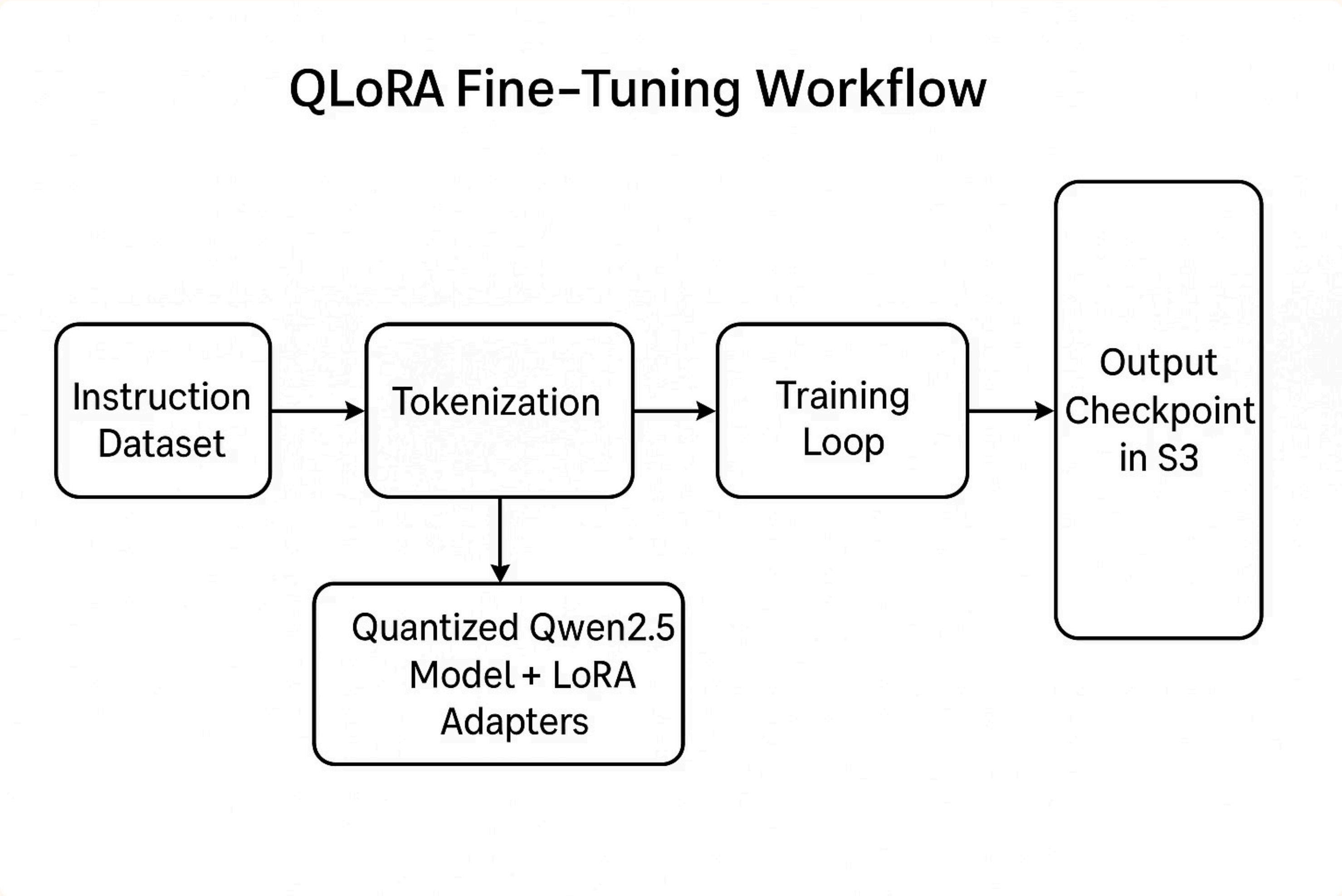
QLoRA enables efficient fine-tuning of large language models on consumer hardware by combining quantization and low-rank adaptation.

To convert Qwen2.5-Coder-7B into an instructional CS tutor, we applied QLoRA:

Training Setup:

- 4-bit NF4 quantization (reduces model size by 75%)
- LoRA rank = 16, α = 32 (updates only ~40M parameters, <1% of total)
- Batch size optimized for RTX 4090 (8 samples per batch)
- Training duration: [X epochs over Y hours]
- Perplexity decreased from [initial] → [final] across training steps
- Checkpoints stored in S3 /checkpoints/ for reproducibility

This approach achieves high pedagogical quality while maintaining minimal hardware cost and training time.



Tutor Prototype & Model Evaluation

Our fine-tuned model demonstrates strong performance across multiple evaluation metrics and real-world tutoring scenarios.

Our tutor provides:

- Structured explanations for algorithmic concepts (recursion, DP, sorting)
- Debugging guidance for student code mistakes with step-by-step reasoning
- Reformulation of difficult topics using multiple explanation strategies
- Consistent real-time responses (<200ms) enabling interactive tutoring

Evaluation Results:

Metric	Result	Interpretation
BLEU	High	Semantic similarity to expert text
ROUGE-L	Strong	Structural coherence
Perplexity	Low	Smooth domain adaptation
Pass@1	High	Functional correctness
Latency	<200ms	Real-time interaction

Metrics include BLEU (semantic similarity), ROUGE-L (structural coherence), Perplexity (domain adaptation), and Pass@1 (functional correctness).

Final Summary & Closing

This semester, we successfully completed the full ISA pipeline: ingestion → transformation → warehouse → monitoring → model evaluation → demo.

This semester, we successfully built and demonstrated:

- A complete cloud-based data pipeline (Airflow orchestration with automated retries)
- A curated, domain-pure CS explanation dataset (5,000 samples from Cosmopedia)
- A QLoRA-fine-tuned AI tutor (Qwen2.5-Coder-7B with <1% parameter updates)
- A functional real-time inference interface (<200ms response time)
- A full ISA pipeline demo (ingestion, cleaning, transformation, warehouse, monitoring, evaluation)

Forward Path:

- Integrate adaptive personalization using student learning profiles
- Expand dataset to 20K+ instructional samples for broader CS coverage
- Implement reinforcement learning from human feedback (RLHF) for improved tutoring behavior
- Incorporate multimodal CS instruction (diagrams, execution traces, code visualization)

This project demonstrates the feasibility of building production-scale AI tutoring systems using modern ML infrastructure and best practices.

Acknowledgments

Thank you for your attention and support throughout this project. We appreciate the opportunity to share our work with you.